



University
of Exeter

COMM113

Deep Learning – Week 12

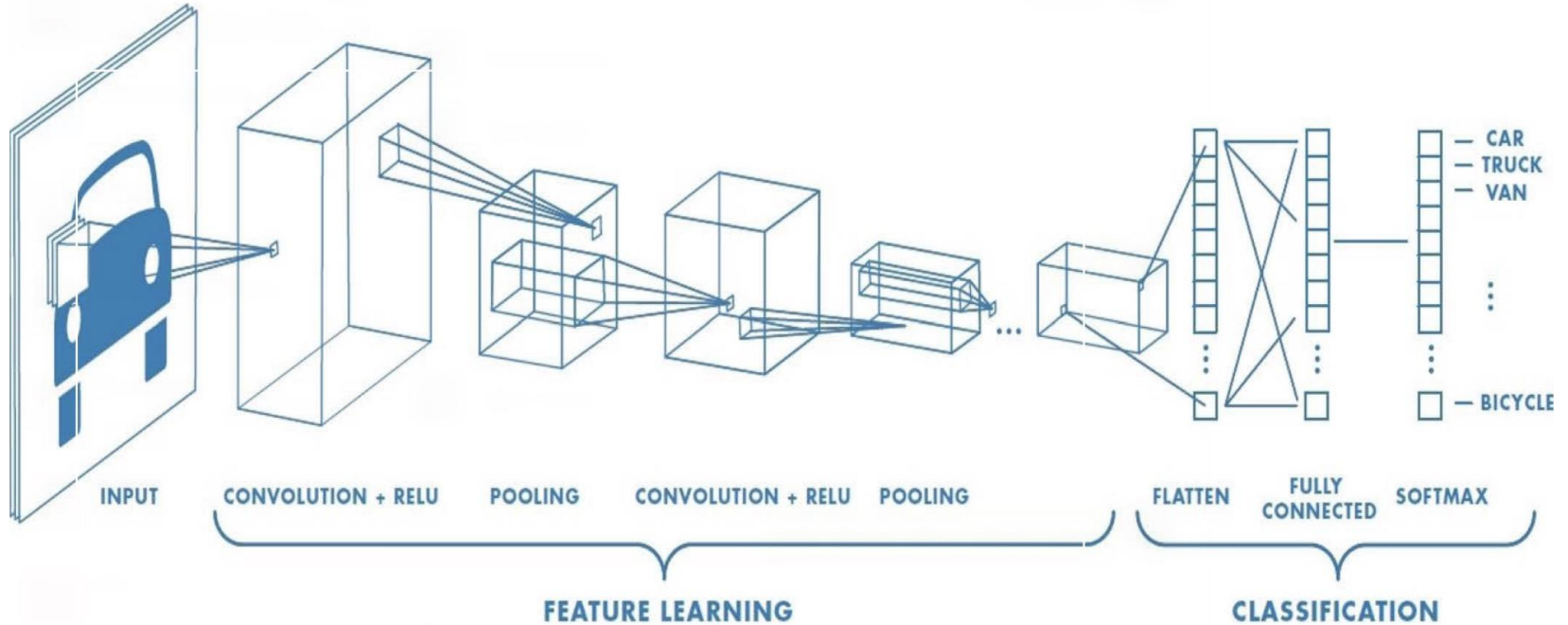
Revision

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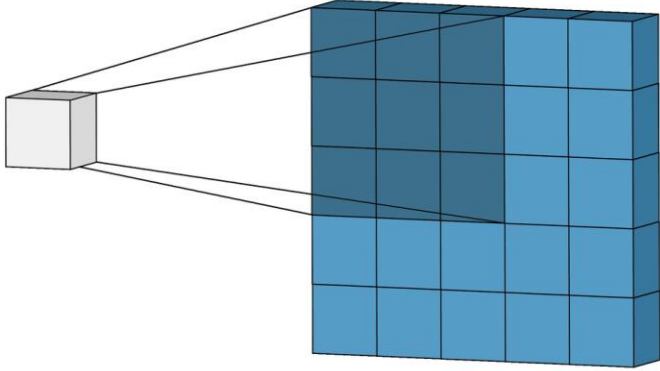
Revision

- CNN
- RNN
- LSTM
- Transformers
- Trustworthy

Convolution Neural Network



Convolution Operation



Slide the filter step by step on the input image.

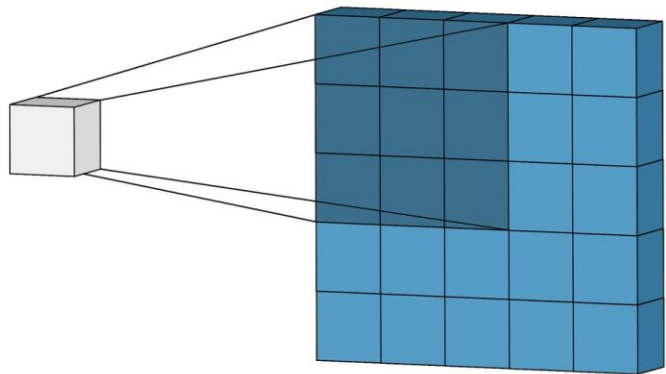
Input 5×5 , Filter $3 \times 3 \rightarrow$

Output: y

Input: x

Filter: w

Convolution Operation



Output: y

Input: x

Filter: w

Slide the filter step by step on the input image.

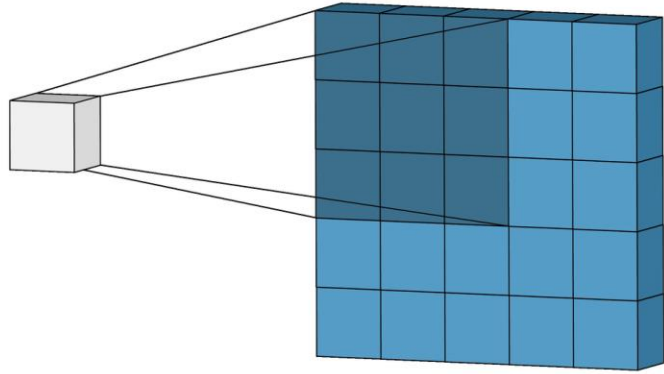
Input 5×5 , Filter $3 \times 3 \rightarrow$

Output 3×3 (no padding, stride = 1)

$$\begin{aligned} y(0,0) = & x(0,0)w(0,0) + x(0,1)w(0,1) + x(0,2)w(0,2) \\ & + x(1,0)w(1,0) + x(1,1)w(1,1) + x(1,2)w(1,2) \\ & + x(2,0)w(2,0) + x(2,1)w(2,1) + x(2,2)w(2,2) \end{aligned}$$

$$\begin{aligned} y(1,1) = & x(1,1)w(0,0) + x(1,2)w(0,1) + x(1,3)w(0,2) \\ & + x(2,1)w(1,0) + x(2,2)w(1,1) + x(2,3)w(1,2) \\ & + x(3,1)w(2,0) + x(3,2)w(2,1) + x(3,3)w(2,2). \end{aligned}$$

Convolution Operation



Output: y

Input: x

Filter: w

Slide the filter step by step on the input image.

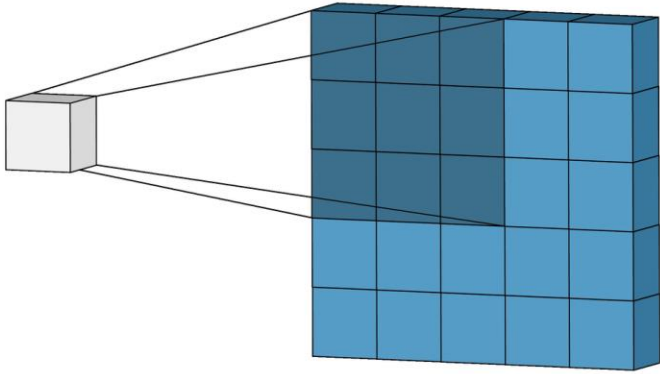
Input 5×5 , Filter $3 \times 3 \rightarrow$

Output 3×3 (no padding, stride = 1)

$$\begin{aligned} y(0,0) = & x(0,0)w(0,0) + x(0,1)w(0,1) + x(0,2)w(0,2) \\ & + x(1,0)w(1,0) + x(1,1)w(1,1) + x(1,2)w(1,2) \\ & + x(2,0)w(2,0) + x(2,1)w(2,1) + x(2,2)w(2,2) \end{aligned}$$

$$y(i,j) = \sum_{m=0}^{k-1} \sum_{n=0}^{k-1} x(i+m, j+n) \cdot w(m,n)$$

Convolution Operation



Stride: how much the filter moves each step.

Input 5×5 , Filter $3 \times 3 \rightarrow$
Output 3×3 (no padding, stride = 1)

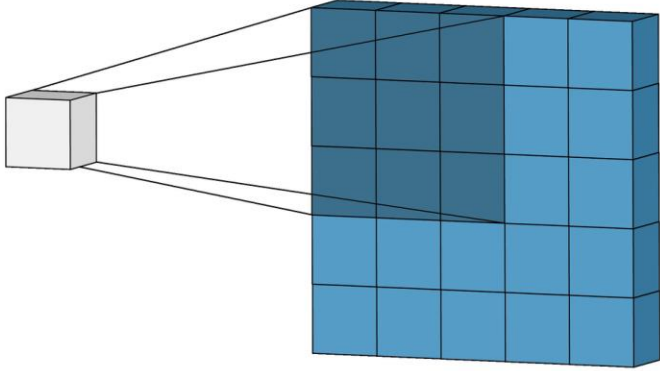
What if stride = 2?

Output: y

Input: x

Filter: w

Convolution Operation



Stride: how much the filter moves each step.

Input 5×5 , Filter $3 \times 3 \rightarrow$
Output 3×3 (no padding, stride = 1)

What if stride = 2?

Input 5×5 , Filter $3 \times 3 \rightarrow$
Output 2×2 (no padding, stride = 2)

Convolution Operation

Padding = 1

0	0	0	0	0	0	0
0	60	113	56	139	85	0
0	73	121	54	84	128	0
0	131	99	70	129	127	0
0	80	57	115	69	134	0
0	104	126	123	95	130	0
0	0	0	0	0	0	0

Kernel

0	-1	0
-1	5	-1
0	-1	0

114				

Zero-Padding: adding zeros to the input

What's the meaning of padding?

Convolution Operation

Padding = 1

0	0	0	0	0	0	0
0	60	113	56	139	85	0
0	73	121	54	84	128	0
0	131	99	70	129	127	0
0	80	57	115	69	134	0
0	104	126	123	95	130	0
0	0	0	0	0	0	0

Kernel

0	-1	0
-1	5	-1
0	-1	0

114				

Padding: adding zeros to the input

What's the meaning of padding?
Can keep the size.

Example - Padding = 1, Stride = 2

After zero-padding (1 pixel border), Input now 7×7

$$x_p = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 1 & 2 & 0 & 0 & 0 \\ 0 & 0 & 2 & 3 & 1 & 0 & 0 \\ 0 & 1 & 2 & 1 & 0 & 0 & 0 \\ 0 & 2 & 1 & 0 & 1 & 1 & 0 \\ 0 & 0 & 1 & 2 & 2 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

$$w = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & -1 \end{bmatrix}$$

Compute by yourself

Example - Padding = 1, Stride = 2

$$x_p = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 1 & 2 & 0 & 0 & 0 \\ 0 & 0 & 2 & 3 & 1 & 0 & 0 \\ 0 & 1 & 2 & 1 & 0 & 0 & 0 \\ 0 & 2 & 1 & 0 & 1 & 1 & 0 \\ 0 & 0 & 1 & 2 & 2 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

$$w = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & -1 \end{bmatrix}$$

$$y(0,0) = 0 \times 1 + 0 \times 0 + 0 \times 1 + 0 \times 0 + 1 \times 1 + 1 \times 0 + 0 \times 1 + 0 \times 0 + 2 \times (-1) = 0 + 0 + 0 + 0 + 1 + 0 + 0 + 0 - 2 = -1$$

$$y(0,1) = 0 \times 1 + 0 \times 0 + 0 \times 1 + 1 \times 0 + 2 \times 1 + 0 \times 0 + 2 \times 1 + 1 \times 0 + 1 \times (-1) = 0 + 0 + 0 + 0 + 2 + 0 + 2 + 0 - 1 = 3$$

$$y(0,2) = 0 \cdot 1 + 0 \cdot 0 + 0 \cdot 1 + 0 \cdot 0 + 0 \cdot 1 + 0 \cdot 0 + 1 \cdot 1 + 0 \cdot 0 + 0 \cdot (-1) = 1$$

Example - Padding = 1, Stride = 2

$$x_p = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 1 & 2 & 0 & 0 & 0 \\ 0 & 0 & 2 & 3 & 1 & 0 & 0 \\ 0 & 1 & 2 & 1 & 0 & 0 & 0 \\ 0 & 2 & 1 & 0 & 1 & 1 & 0 \\ 0 & 0 & 1 & 2 & 2 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

$$y(1,0)=2$$

$$y(1,1)=4$$

$$y(1,2)=2$$

$$w = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & -1 \end{bmatrix}$$

Example - Padding = 1, Stride = 2

$$x_p = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 1 & 2 & 0 & 0 & 0 \\ 0 & 0 & 2 & 3 & 1 & 0 & 0 \\ 0 & 1 & 2 & 1 & 0 & 0 & 0 \\ 0 & 2 & 1 & 0 & 1 & 1 & 0 \\ 0 & 0 & 1 & 2 & 2 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

$$y(2,0)=1$$

What effect does stride=2 have?

$$y(2,1)=4$$

$$y(2,2)=1$$

$$w = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & -1 \end{bmatrix}$$

Pooling Layer

- Spatial dimension reduction
- Local invariance
- No parameters: max or average of 2x2 (or nxn) units

1	1	2	4
5	6	7	8
3	2	1	0
1	2	3	4

max pool with 2x2 filters
and stride 2



6	8
3	4

$$(4-2)/2 + 1 = 2$$

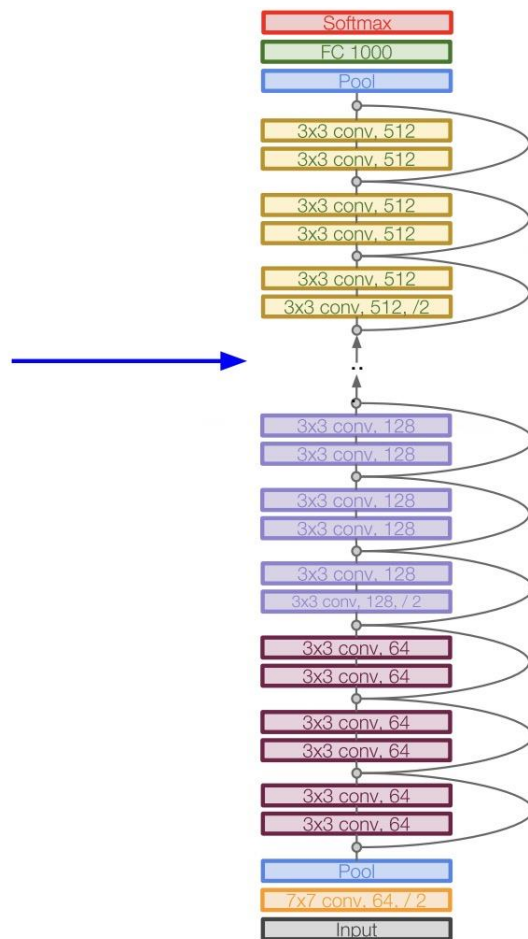
$$(\text{Input size} - \text{filter size})/\text{stride} + 1$$

ResNet

[He et al., 2015]

Total depths of 34, 50, 101, or
152 layers for ImageNet

What problem do residual connections solve?



ResNet

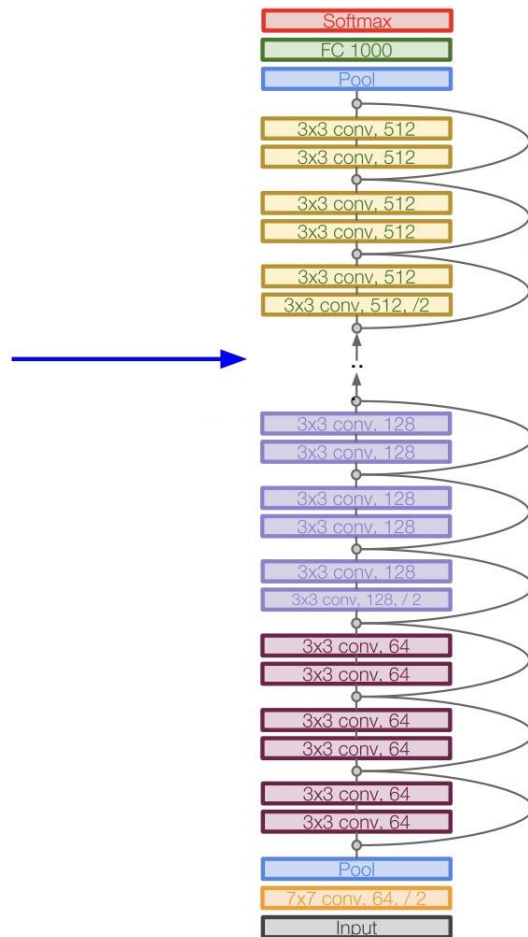
[He et al., 2015]

Total depths of 34, 50, 101, or 152 layers for ImageNet

What problem do residual connections solve?

Vanishing gradients in deep nets.

How do skip connections affect optimisation?



Computer Vision Tasks

Classification



CAT

No spatial extent

Semantic Segmentation



**GRASS, CAT,
TREE, SKY**

No objects, just pixels

Object Detection



DOG, DOG, CAT

Multiple Object

Instance Segmentation



DOG, DOG, CAT

[This image is CC0 public domain](#)

RNNs

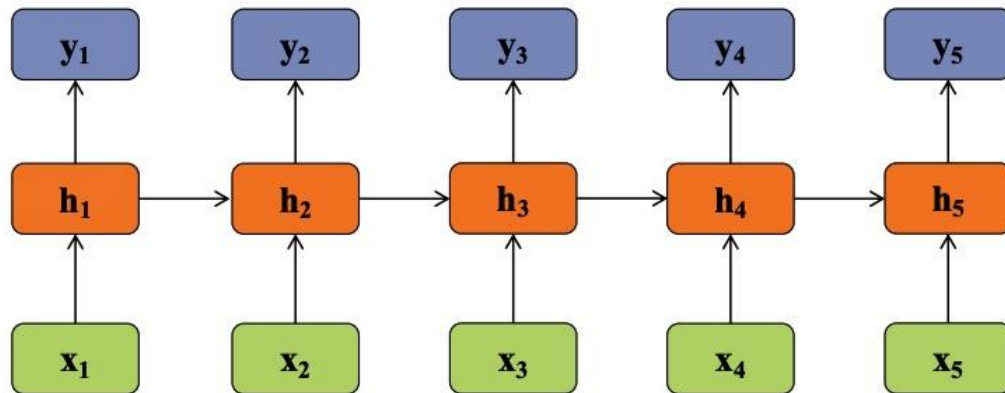
inputs: $\mathbf{x} = (x_1, x_2, \dots, x_T), x_i \in \mathcal{R}^I$
hidden units: $\mathbf{h} = (h_1, h_2, \dots, h_T), h_i \in \mathcal{R}^J$
outputs: $\mathbf{y} = (y_1, y_2, \dots, y_T), y_i \in \mathcal{R}^K$
nonlinearity: $\mathcal{H}$

Definition of the RNN:

$$h_t = \mathcal{H}(W_{xh}x_t + W_{hh}h_{t-1} + b_h)$$

$$y_t = W_{hy}h_t + b_y$$

An RNN is a neural network designed to process sequences. It maintains a *hidden state* that carries information from past timesteps.



RNNs

inputs: $\mathbf{x} = (x_1, x_2, \dots, x_T), x_i \in \mathcal{R}^I$
hidden units: $\mathbf{h} = (h_1, h_2, \dots, h_T), h_i \in \mathcal{R}^J$
outputs: $\mathbf{y} = (y_1, y_2, \dots, y_T), y_i \in \mathcal{R}^K$
nonlinearity: $\mathcal{H}$

Definition of the RNN:

$$h_t = \mathcal{H}(W_{xh}x_t + W_{hh}h_{t-1} + b_h)$$

$$y_t = W_{hy}h_t + b_y$$

- If $T=1$, then we have a standard feed-forward neural net with one hidden layer, which requires fixed size inputs/outputs
- By contrast, an RNN can handle arbitrary length inputs/outputs because T can vary from example to example
- The **key idea is that we reuse the same parameters** at every timestep, always building off of the previous hidden state

Long Short-Term Memory (LSTM)

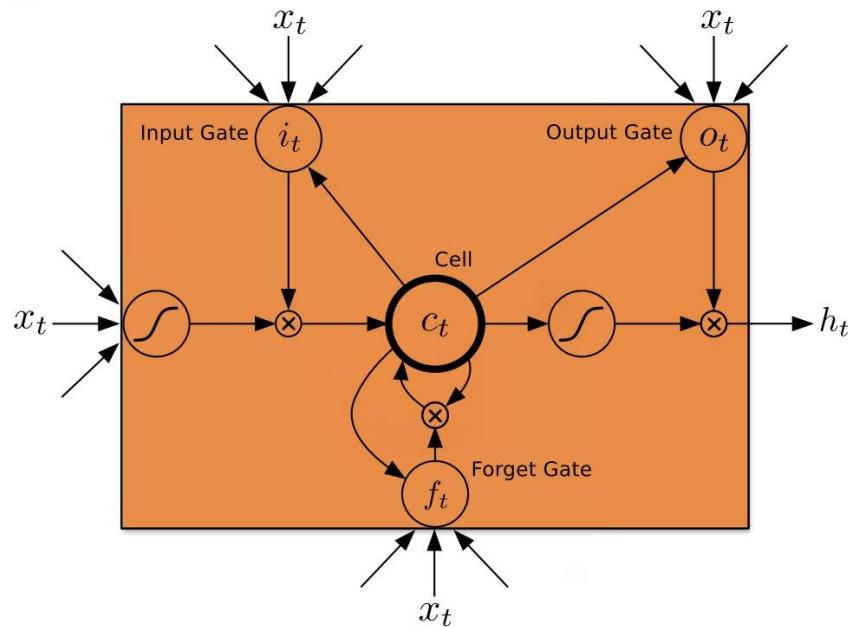
Input gate: masks out the standard RNN inputs

Forget gate: masks out the previous cell

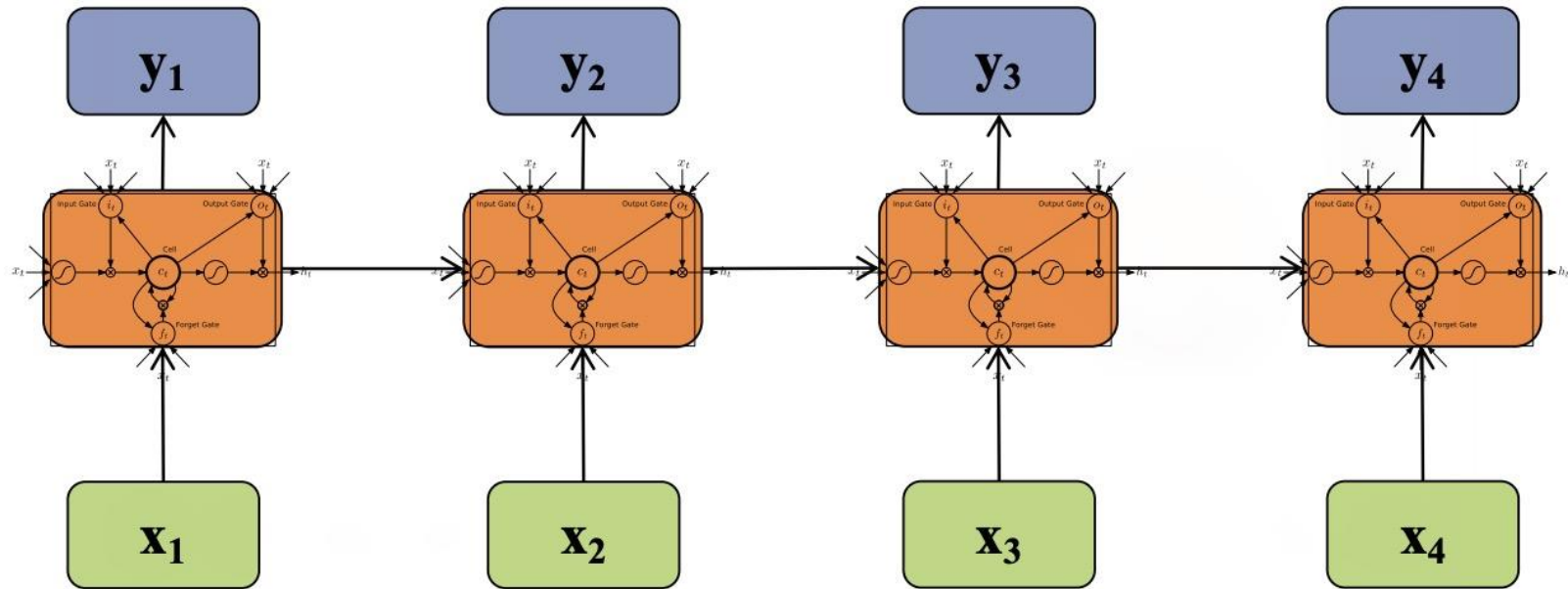
Cell: stores the input/forget mixture

Output gate: masks out the values of the next hidden

$$\begin{aligned}i_t &= \sigma(W_{xi}x_t + W_{hi}h_{t-1} + W_{ci}c_{t-1} + b_i) \\f_t &= \sigma(W_{xf}x_t + W_{hf}h_{t-1} + W_{cf}c_{t-1} + b_f) \\c_t &= f_t c_{t-1} + i_t \tanh(W_{xc}x_t + W_{hc}h_{t-1} + b_c) \\o_t &= \sigma(W_{xo}x_t + W_{ho}h_{t-1} + W_{co}c_t + b_o) \\h_t &= o_t \tanh(c_t)\end{aligned}$$



Long Short-Term Memory (LSTM)



Self-Attention

Step 1: With embeddings stacked in X , calculate **queries**, **keys**, and **values**.

$$Q = XW^Q \quad K = XW^K \quad V = XW^V$$

Step 2: Calculate attention score between **query** and **keys**.

$$E = QK^T$$

Step 3: Take the softmax to normalise attention scores.

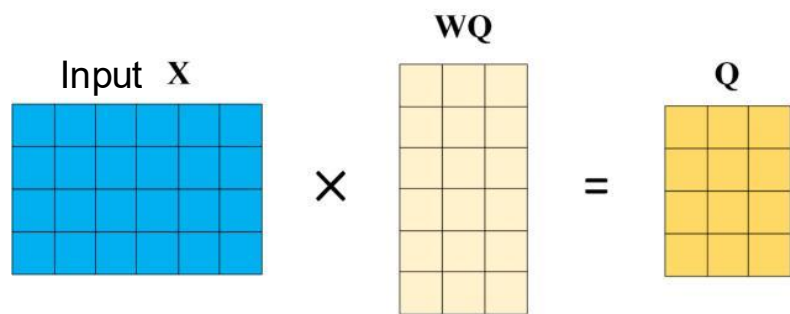
$$A = \text{softmax}(E)$$

Step 4: Take a weighted sum of **values**.

$$\text{Output} = AV$$

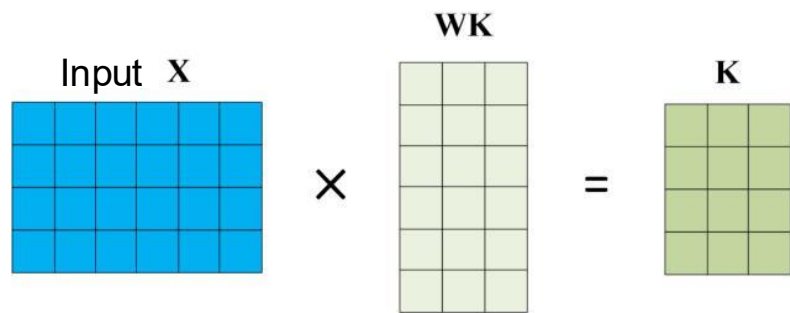
$$\text{Output} = \text{softmax}(QK^T)V$$

Input X $\times$ WQ $=$ Q



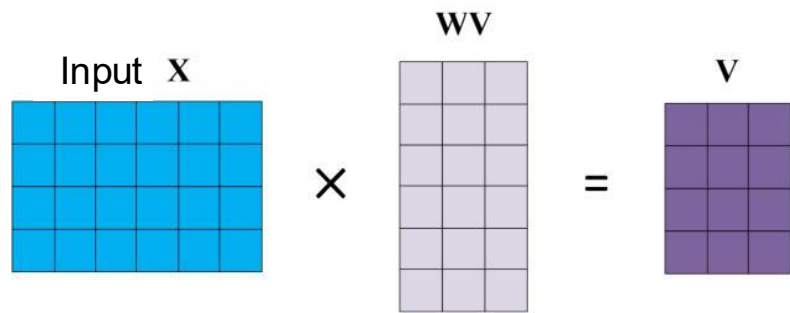
The diagram illustrates the matrix multiplication of Input X (a 4x8 blue grid) by WQ (an 8x4 yellow grid) to produce the output Q (a 4x4 yellow grid). The multiplication is represented by the equation $X \times WQ = Q$.

Input X $\times$ WK $=$ K

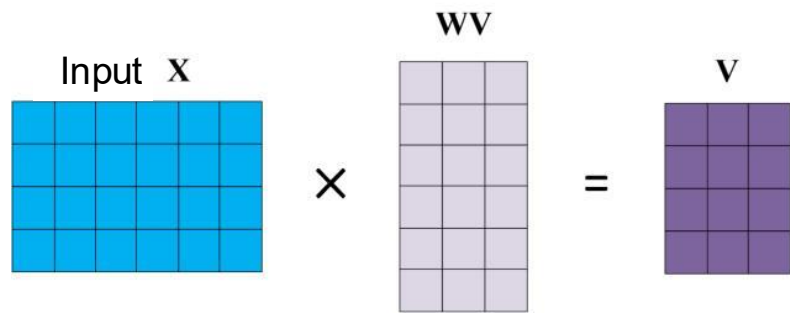
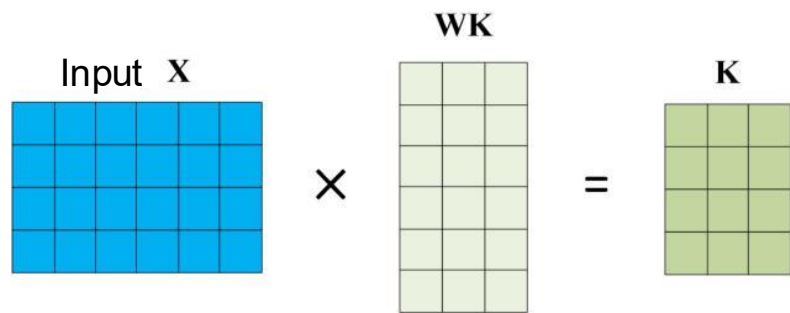
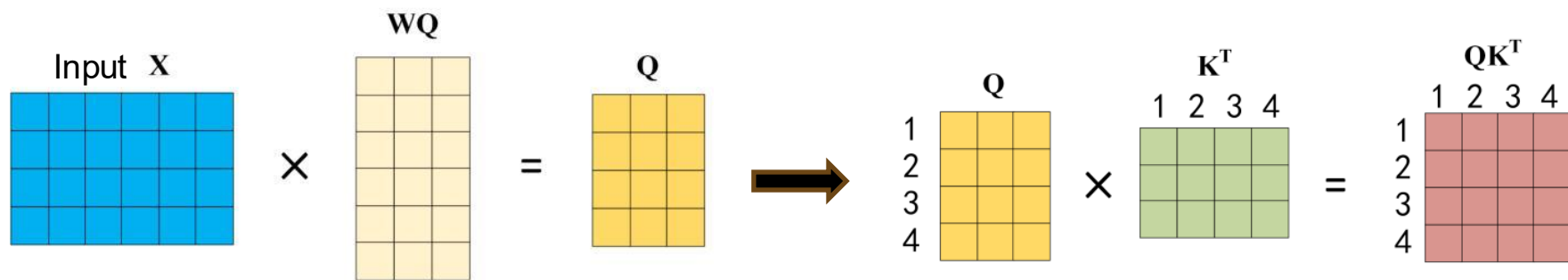


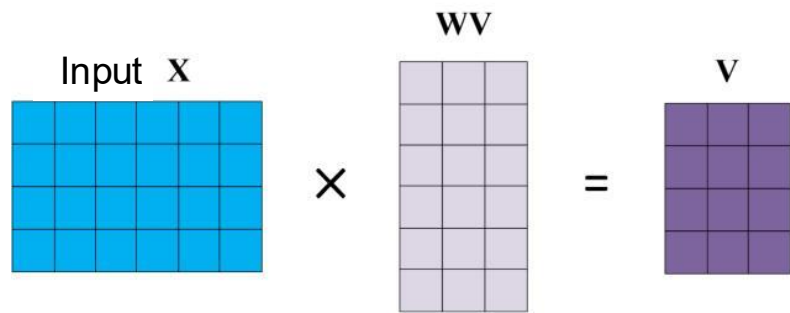
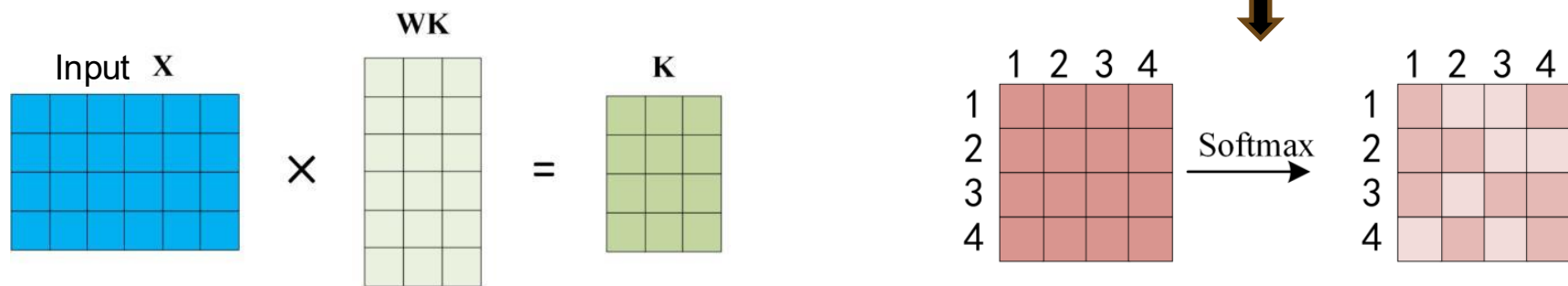
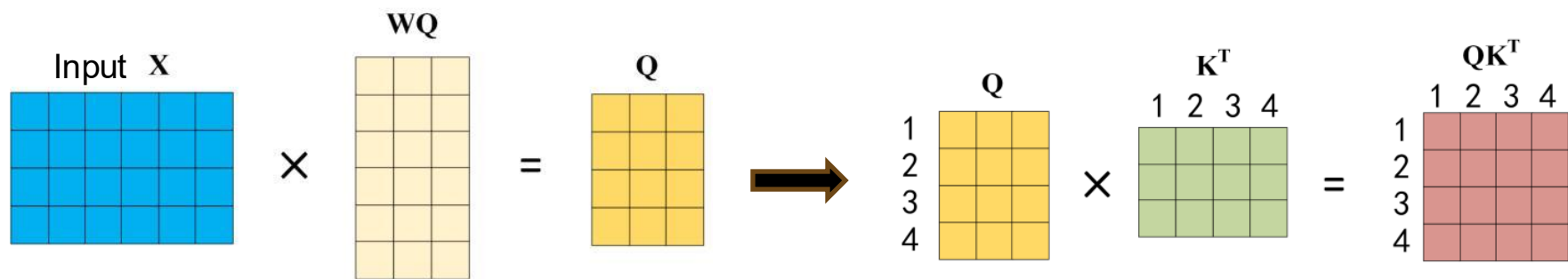
The diagram illustrates the matrix multiplication of Input X (a 4x8 blue grid) by WK (an 8x4 light green grid) to produce the output K (a 4x4 light green grid). The multiplication is represented by the equation $X \times WK = K$.

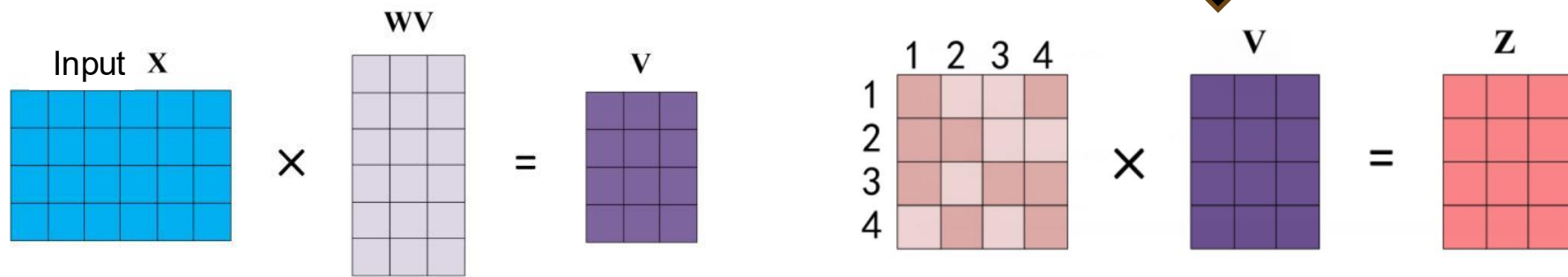
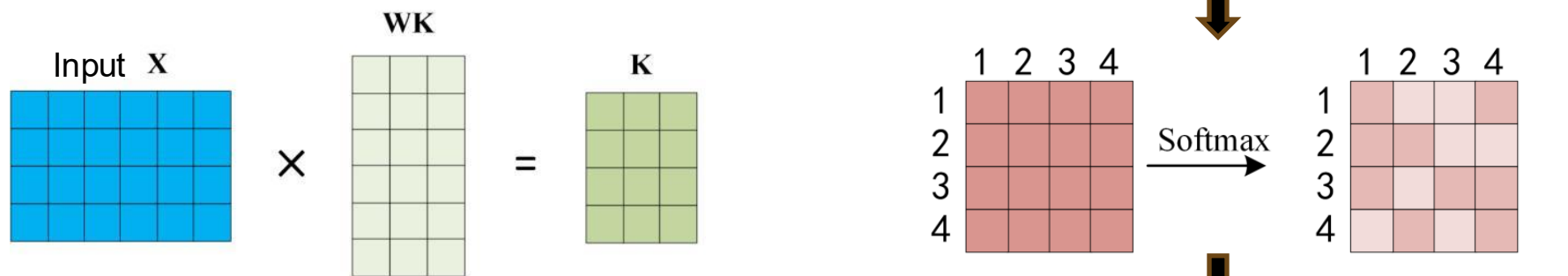
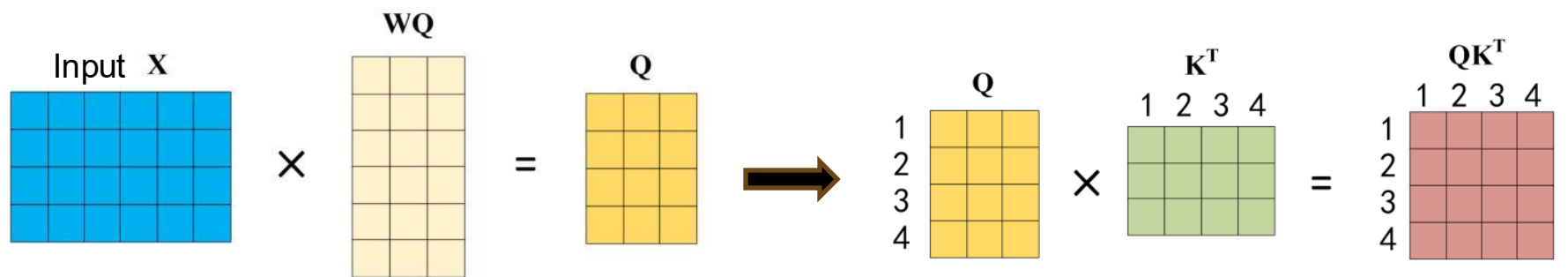
Input X $\times$ WV $=$ V



The diagram illustrates the matrix multiplication of Input X (a 4x8 blue grid) by WV (an 8x4 light purple grid) to produce the output V (a 4x4 dark purple grid). The multiplication is represented by the equation $X \times WV = V$.

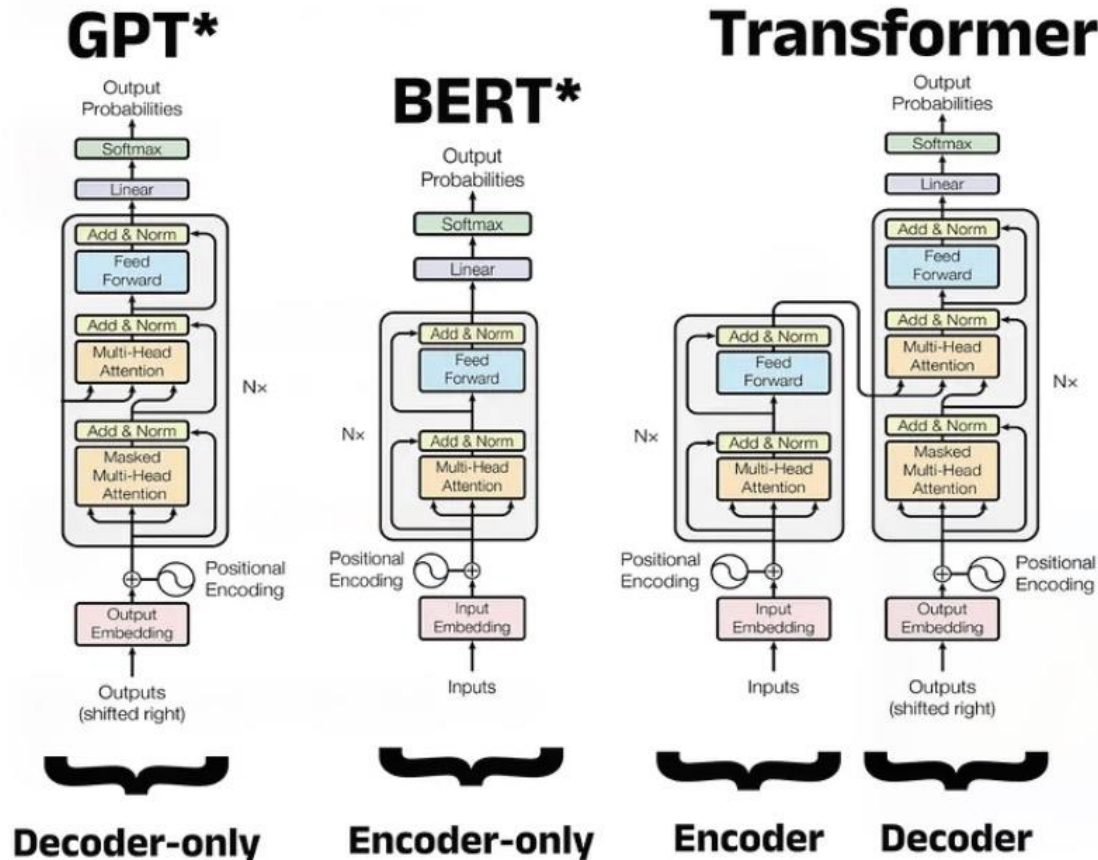






GPT - BERT - Transformer

GPT predicts the next word based only on the words before it. This is why GPT is excellent at text generation.



BERT sees the entire sentence at once. It **understands** context in both directions.

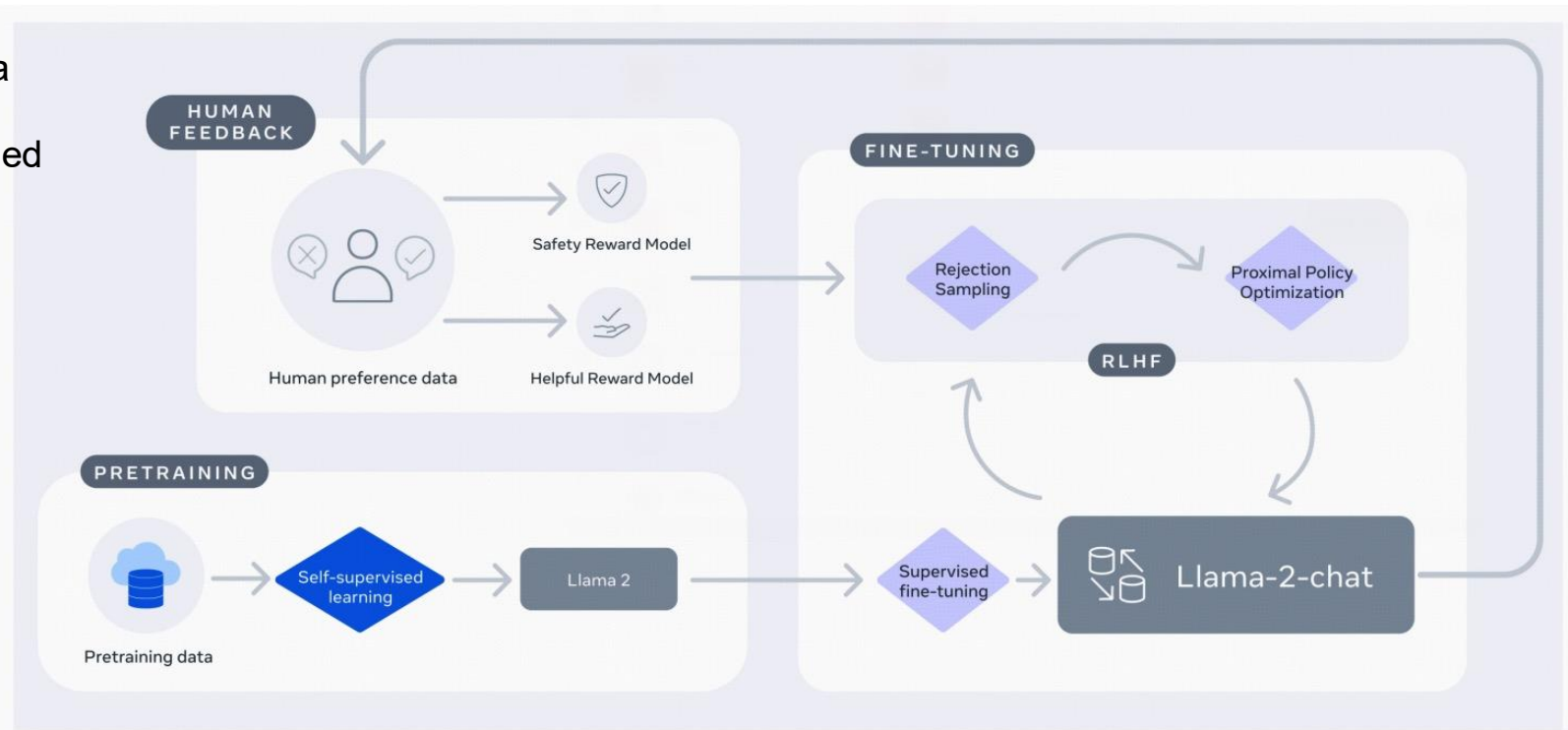
Training LLMs - Llama as an example



Pretraining =
massive raw data

SFT = task-focused
correction

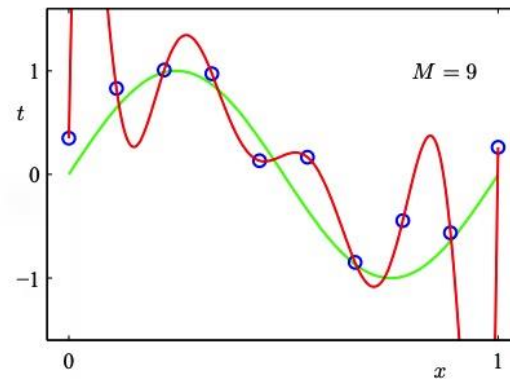
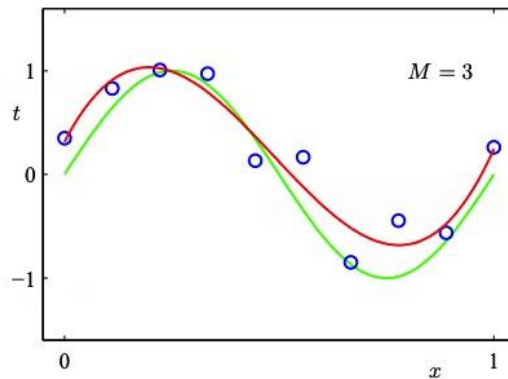
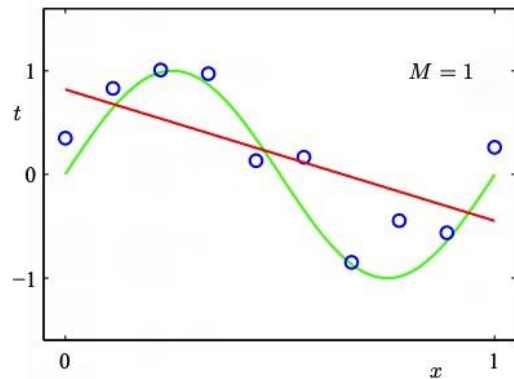
RLHF = human
alignment



Pre-training -> Supervised Fine-tuning (SFT) -> RLHF Pre-training -> Post-training
Pre-training -> Mid-training -> Post-training

Generalisation

- Recall: overfitting and underfitting

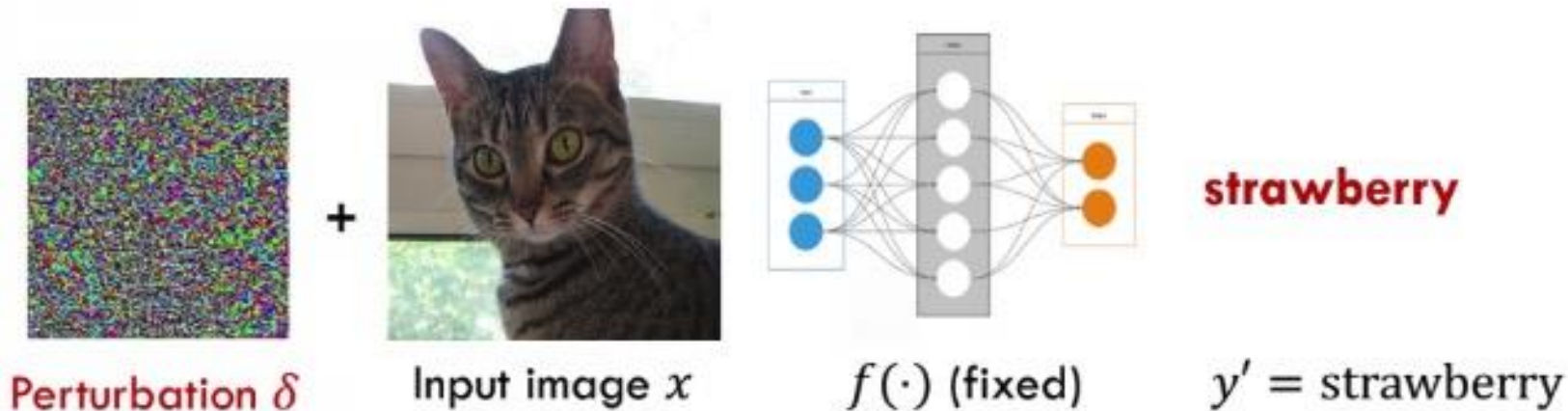


We'd like to minimise the generalisation error, i.e. error on new examples.

Adversarial Robustness

- A motivation example

Given a well-trained classification model for ImageNet. How to **alter the model output by changing the input?**



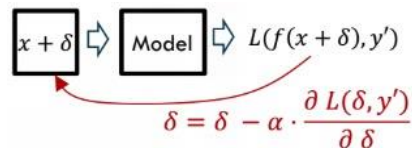
1. Rephrase what we want

Find δ such that $f(x + \delta) = y' \neq y$
For example, $y' = \text{strawberry}$

2. Defining the loss function

$$\min_{\delta} L(f(x + \delta), y')$$

3. Optimise the perturbation


$$\delta = \delta - \alpha \cdot \frac{\partial L(\delta, y')}{\partial \delta}$$

Fairness

Models inherit patterns from data. If data embeds historical inequities, neural networks amplify them.

Unfair Examples:

- COMPAS recidivism case: higher false-positive rates for Black defendants.
- Amazon hiring model: penalised CVs containing “women’s”.
- Image classifiers (e.g., classifying doctors as men, nurses as women).

Interpretability

What does interpretation look like?

- Heatmap Visualisation



Figure 3. Attribution for Diabetic Retinopathy grade prediction from a retinal fundus image. The original image is shown on the left, and the attributions (overlayed on the original image in grayscale) is shown on the right. On the original image we annotate lesions visible to a human, and confirm that the attributions indeed point to them.

[Sundarajan 2017]

in a clinical trial mainly involving patients over qqq with coronary heart disease , ramipril reduced mortality while vitamin e had no preventive effect .

in a clinical trial mainly involving patients over qqq with coronary heart disease , ramipril reduced mortality while vitamin e had no preventive effect .

in a clinical trial mainly involving patients over qqq with coronary heart disease , ramipril reduced mortality while vitamin e had no preventive effect .

Table 2: Gate activations for each aspect in a PICC abstract. Note that because gates are calculated at the final convolution layer, activations are not in exact 1-1 correspondence with words.

[Jain 2018]

Thank you for your effort this term!

Have a wonderful Christmas break 🎁

Exam: May 2026